

Real-Time Driver Behavior Detection Using Lightweight Object Detection Models with Subject-Disjoint Evaluation

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Abstract—Driver Monitoring Systems (DMS) are critical safety components of modern Advanced Driver Assistance Systems (ADAS). While commercial systems often employ complex temporal pipelines, automotive edge deployments demand lightweight, low-latency 2D object detectors capable of real-time single-frame inference. This paper presents DMS-Eval, an empirical benchmark comparing compact convolutional detectors (Ultralytics YOLO11n and YOLO26n) against lightweight Real-Time Detection Transformers (D-FINE-N). Formulated on 81 naturalistic in-cabin RGB video sequences from the Driver Monitoring Dataset (DMD) across 14 subjects, our benchmark establishes a standardized dataset of 15,723 frames (640×640 crop). Ground truth is constructed via 100% manual human annotation in Label Studio targeting four visual warning cues: `phone_use`, `drinking`, `yawning`, and `hand_over_mouth`, incorporating 12,722 naturalistic negative frames (80.91%) to enforce false-positive suppression. To eliminate identity leakage, we formalize an authoritative 8/3/3 subject-disjoint split via deterministic exhaustive combinatorial search across 60,060 candidate assignments, ensuring all splits preserve identical class proportions ($\leq 5.48\%$ relative divergence). Under a strictly controlled comparison on dedicated RTX 4060 hardware at batch size 1 across 220 fine-tuning epochs, candidate models are calibrated via validation-only F_1 score maximization prior to single-pass test evaluation. This work provides an open, reproducible framework for edge-deployed in-cabin driver state analysis.

Index Terms—driver monitoring systems, driver drowsiness detection, driver distraction detection, lightweight object detection, computer vision, YOLO, DETR, subject-disjoint evaluation, real-time edge AI

I. INTRODUCTION

In-cabin Driver Monitoring Systems (DMS) constitute a vital component of modern Advanced Driver Assistance Systems (ADAS) and autonomous vehicle safety suites, aiming to prevent roadway collisions through early visual detection of driver drowsiness and inattention [18], [19]. While high-level driver state assessment often leverages temporal recurrent or multi-frame sequence models, automotive edge deployments impose severe compute, memory, and latency constraints. Consequently, there is an urgent practical demand for lightweight, nano-scale 2D object detectors capable of executing real-time single-frame inference on embedded in-cabin hardware.

To systematically evaluate modern compact detector paradigms under strictly controlled single-frame constraints, this paper introduces **DMS-Eval**, an empirical benchmark comparing real-time nano-scale object detectors. We evaluate leading single-stage convolutional detectors (Ultralytics YOLO11n [1] and Ultralytics YOLO26n [2]) against modern end-to-end Real-Time Detection Transformers (D-FINE-N [3]).

The benchmark is developed on real-cabin RGB video from the Driver Monitoring Dataset (DMD) [8], incorporating all three original dataset session categories: *distraction*, *drowsiness*, and *gaze*. Across 81 driver-facing videos from 14 volunteer participants, frames are sampled uniformly at 1 FPS and cropped to a frozen 640×640 spatial window ($x = 272, y = 71, w = 640, h = 640$) as illustrated in Fig. 1, yielding a standardized dataset of 15,723 frames. Retaining all three session folders—particularly the *gaze* recordings and non-cue intervals—provides an abundant volume of naturalistic negative background frames (0 bounding boxes), preventing compact detectors from overtraining on positive targets and enforcing realistic false-positive suppression during routine driving (illustrated in Fig. 3a). Ground truth is constructed through 100% direct manual expert annotation in Label Studio [20] with format exports structured via `label-studio-converter` [21] under the Microsoft COCO format [11] targeting 4 frozen visual warning cues (exemplified in Fig. 2 and quantified in Fig. 3b):

- 1) `yawning`: mouth region only (wide oral opening/distension);
- 2) `hand_over_mouth`: full head/face enclosing the occluding hand;
- 3) `drinking`: interacting hand and beverage container together in active consumption posture;
- 4) `phone_use`: handheld phone and interacting hand held to ear/head in calling posture.

Under our single-frame annotation protocol, each sampled image contains at most one visual warning cue annotation (0 or 1 bounding box per image). In particular, *yawning*

and `hand_over_mouth` are mutually exclusive; if a driver yawns while a hand visibly occludes the mouth, the instance is uniquely labeled as `hand_over_mouth`, ensuring non-overlapping ground-truth targets.

All models are evaluated under a strict 8/3/3 subject-disjoint split ($S_{\text{train}} \cap S_{\text{val}} = \emptyset$, $S_{\text{train}} \cap S_{\text{test}} = \emptyset$, $S_{\text{val}} \cap S_{\text{test}} = \emptyset$) selected via exhaustive combinatorial search across 60,060 candidate assignments ($\leq 5.48\%$ maximum distribution divergence). Training is executed under identical hardware (NVIDIA RTX 4060), batch size 1, 220 epochs, FP16 AMP, with optimal thresholds selected strictly on the validation split prior to isolated test evaluation.

A. Research Gap

Despite substantial progress in driver monitoring, a critical methodological gap persists: **no existing study provides a controlled, head-to-head comparison between nano-scale convolutional detectors and nano-scale Real-Time Detection Transformers for in-cabin driver cue localization under identical hardware, data, and training constraints.** Existing work leaves this gap unaddressed along four axes:

(i) *Classification without spatial localization.* The AUC benchmark [9] and ICK-PANet [13] classify full images into posture categories but produce no bounding boxes localizing the specific objects or facial cues. For edge deployment, spatial localization is essential to support downstream tracking and alert subsystems. (ii) *Multi-stage pipelines with compounding latency.* Recent DMD studies [14] chain YOLO11, AlphaPose, and LSTM networks, introducing multi-model penalties that preclude real-time edge execution. Video-sequence benchmarks [10], [12] require frame buffers and recurrent state unsuitable for single-frame inference. (iii) *Single-family evaluations.* Lightweight CNN detectors [6], [7] and real-time DETRs [3], [4] are benchmarked exclusively on COCO [11] and never compared against each other within the in-cabin DMS domain. (iv) *Absent evaluation rigor.* Most YOLO-based DMS studies use random frame-level splits, enabling identity leakage, and report inference speed as theoretical GFLOPs or large-batch throughput rather than the batch-size-1 latency that governs edge responsiveness [15], [16].

All three candidates evaluated here operate below 5M parameters (YOLO11n: 2.6M, YOLO26n: 2.4M, D-FINE-N: 3.8M), fitting within the on-chip SRAM envelope of automotive-grade microcontrollers [17]. Because GFLOPs diverge from measured latency due to memory-bound attention, non-fused depthwise convolutions, and NMS overhead [16], this benchmark will measure wall-clock latency via CUDA event synchronization at batch size 1 following MLPerf guidelines [15].

B. Key Contributions

To close the identified gap, this paper will make four contributions:

- **First Cross-Paradigm Nano-Scale DMS Benchmark:** The first controlled comparison between nano-scale CNNs (YOLO11n, YOLO26n) and a nano-scale DETR

(D-FINE-N) for in-cabin driver cue detection with spatial bounding box localization.

- **Rigorous Subject-Disjoint Data Curation:** 15,723 frames (640×640) with 100% manual annotation across 4 warning cues and 80.91% naturalistic negatives, partitioned via exhaustive combinatorial optimization ($\leq 5.48\%$ divergence).
- **Unified Precision-Efficiency Protocol:** Identical hardware (RTX 4060), batch size 1, 220 epochs, FP16 AMP, with detection accuracy and hardware-synchronized latency ($p50/p95/p99$) reported jointly.
- **Open Reproducibility:** Public release of all COCO annotations, partitioning scripts, and evaluation tooling.

C. Research Question and Scope

This study targets **frame-level observable cues of driver drowsiness and distraction.** Head orientation behaviors are excluded: in isolated 1 FPS frames without temporal context, safe mirror glances cannot be distinguished from dangerous inattention. The benchmark focuses on four unambiguous visual warning cues.

RQ: *Under a controlled, compute-constrained evaluation, how do sub-5M-parameter detectors from fundamentally different paradigms—convolutional single-stage detectors versus Real-Time Detection Transformers—compare in detection accuracy, inference latency, and false-alarm suppression when identifying spatial visual cues of driver distraction and drowsiness on edge hardware?*

II. RELATED WORK

A. Vision-Based Driver Monitoring Systems

Early DMS literature focused on geometric facial landmark analysis using heuristic indicators such as Eye Aspect Ratio (EAR), PERCLOS, and Mouth Aspect Ratio (MAR) [19]. In-cabin distraction benchmarks—including the AUC Distracted Driver dataset [9], Drive&Act [10], and UTA-RLDD [12]—advanced behavioral understanding but rely on full-image classification or video-sequence models without spatial bounding box localization. The specific limitations of these approaches and subsequent work are analyzed in Section I-A.

B. Real-Time Lightweight Object Detectors

Convolutional detectors have evolved rapidly: YOLO11 uses C3k2 and SPPF modules [1]; YOLO26 eliminates NMS via end-to-end anchor-free design [2]; YOLOv10 introduces dual-label assignment [6]; and YOLOv12 incorporates attention-centric mechanisms [7]. In parallel, RT-DETR [4] adapted the DETR set-prediction paradigm [5] to real-time execution, and D-FINE [3] refined bounding box regression via Fine-grained Distribution Refinement (FDR). However, these architectures are benchmarked on general-purpose datasets and have not been compared head-to-head within the in-cabin DMS domain.

C. Subject Leakage and Evaluation Rigor

Random train/test partitioning across the same human subjects allows models to memorize biometric features rather than visual cues, inflating accuracy [18]. DMS-Eval enforces subject-disjoint partitioning ($S_{\text{train}} \cap S_{\text{test}} = \emptyset$).

III. DATASET FORMULATION AND ANNOTATION PROTOCOL

A. Source Video Corpus and Spatial Standardization

The benchmark is constructed from the Driver Monitoring Dataset (DMD) [8], recorded in real vehicle cabins under variable daylight and ambient lighting conditions. The dataset spans 81 driver-facing RGB videos recorded at ≈ 29.76 FPS across 14 diverse human subjects (S_1 through S_{14}).

To eliminate background noise outside the driver’s operational workspace while preserving high spatial fidelity for fine-grained cues, every frame is uniformly sampled at 1 FPS and cropped to a standardized 640×640 pixel spatial bounding window ($x = 272, y = 71, w = 640, h = 640$) as shown in Fig. 1. This produces exactly 15,723 standardized images.

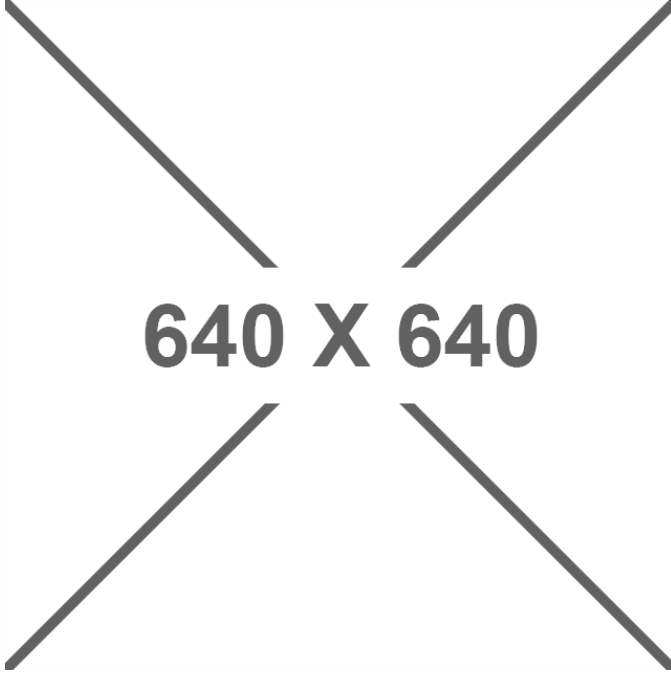


Fig. 1. In-cabin driver-facing frame extraction on DMD video with standardized 640×640 spatial window cropping ($x = 272, y = 71, w = 640, h = 640$), centering the driver’s head, face, and torso for low-latency single-frame object detection.

B. Target Cue Ontology and Annotation Constraints

Ground truth is generated through 100% manual human inspection in Label Studio [20] with format conversion managed via `label-studio-converter` [21]. Bounding box boundaries are defined with strict anatomical extents:

- **yawning** (ID 1): Encloses only the active mouth aperture during wide distension. Head and cheeks are ex-

cluded to prevent confusing normal open-mouth speech with yawning.

- **hand_over_mouth** (ID 2): Encloses the full head and face including the occluding hand when placed over the mouth area.
- **drinking** (ID 3): Encloses the interacting hand and the beverage container together during active consumption posture.
- **phone_use** (ID 4): Encloses the handheld device and interacting hand held to the ear/head in calling posture.



Fig. 2. Visual demonstrations of our manual annotations in Label Studio across all 4 frozen target warning cues: (a) yawning (top-left, mouth aperture only), (b) hand_over_mouth (top-right, full head enclosing occluding hand), (c) drinking (bottom-left, hand and beverage container in consumption posture), and (d) phone_use (bottom-right, hand and mobile device held at the ear in calling posture).

C. Single-Annotation and Mutual Exclusivity Protocol

Under our single-frame protocol, each sampled frame contains at most one bounding box (0 or 1 box). In particular, yawning and hand_over_mouth are strictly mutually exclusive. If a driver yawns while a hand visibly occludes the mouth, the frame is uniquely labeled as hand_over_mouth. This eliminates ambiguous overlapping bounding boxes and ensures sharp ground-truth targets.

D. Negative Sample Richness

By incorporating all three DMD session folders, our benchmark contains 12,722 true negative background frames (80.91%) and 3,001 positive cue frames (19.09%) as shown in Fig. 3.

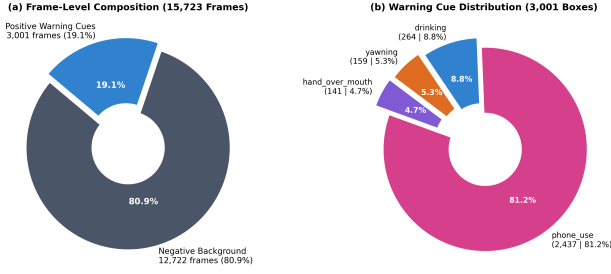


Fig. 3. DMS-Eval dataset and annotation distributions: (a) frame-level dataset composition; (b) proportion of bounding boxes across the 4 frozen target warning cues.

IV. METHODOLOGY AND SUBJECT-DISJOINT PARTITIONING

A. Subject-Disjoint Splitting Principle

To eliminate identity leakage, the 14 subjects are partitioned into 8 Training, 3 Validation, and 3 Test subjects such that:

$$S_{\text{train}} \cap S_{\text{val}} = \emptyset, \quad S_{\text{train}} \cap S_{\text{test}} = \emptyset, \quad S_{\text{val}} \cap S_{\text{test}} = \emptyset \quad (1)$$

B. Exhaustive Proportion-Matching Optimization

Because subjects naturally exhibit varying behavioral frequencies, random assignment often results in severe class imbalance. We formalize an authoritative selection rule:

Select the 8/3/3 subject split whose negative/positive frame proportion and four class proportions most closely match the complete dataset distribution.

Our algorithm exhaustively evaluates all $\binom{14}{8} \times \binom{6}{3} = 3003 \times 20 = 60,060$ ordered 8/3/3 subject assignments. For each candidate partition $\mathcal{P} \in \Omega$ and each split $s \in \{\text{train}, \text{val}, \text{test}\}$, we compute the absolute relative deviation across 5 quantities $\mathcal{Q} = \{\text{pos_rate}, \text{phone}, \text{drink}, \text{yawn}, \text{hand_mouth}\}$:

$$\text{Dev}(q, s) = \frac{|V(q, s) - V_{\text{global}}(q)|}{V_{\text{global}}(q)} \quad (2)$$

Candidate partitions are ranked via a deterministic lexicographic objective:

$$\mathcal{P}^* = \arg \min_{\mathcal{P} \in \Omega} \left(\max_{q \in \mathcal{Q}, s \in \mathcal{S}} \text{Dev}(q, s), \text{RMSE}(\mathcal{P}), \text{Dev}_{\text{test}}(\mathcal{P}) \right) \quad (3)$$

C. Frozen Split Statistics

The optimal partition selected by the algorithm is permanently frozen in `dataset/splits.json`:

- **Train** (8 subjects): $S_1, S_4, S_6, S_7, S_8, S_9, S_{13}, S_{14}$
- **Validation** (3 subjects): S_2, S_3, S_{11}
- **Test** (3 subjects): S_5, S_{10}, S_{12}

As documented in Table I and Table II, the maximum relative deviation across all splits and classes is restricted to $\leq 5.4812\%$, ensuring near-identical distribution fidelity across partitions.

D. Temporal Shuffling and Sequence Preservation

Because 1 FPS extraction from continuous driving video introduces temporal correlation between adjacent frames, a deterministic pseudo-random permutation (seed 13) is applied exclusively to the training split (S_{train}) to diversify mini-batch gradient updates and prevent sequential overfitting. Conversely, validation and test splits ($S_{\text{val}}, S_{\text{test}}$) are strictly preserved in their native chronological order to enable contiguous temporal error analysis and ensure benchmark compatibility with future sequence-based architectures.

TABLE I
FRAME COMPOSITION ACROSS FROZEN BENCHMARK SPLITS

Split	Total Images	Negatives (0 box)	Positives (1 box)	Pos. Rate (%)
Global	15,723	12,722	3,001	19.09%
Train	9,087	7,339	1,748	19.24%
Val	3,423	2,784	639	18.67%
Test	3,213	2,599	614	19.11%

TABLE II
TARGET WARNING CUE COUNTS AND PROPORTIONS ACROSS SPLITS

Split	phone_use	drinking	yawning	hand_mouth
Global	2,437 (81.21%)	264 (8.80%)	159 (5.30%)	141 (4.70%)
Train	1,417 (81.06%)	154 (8.81%)	94 (5.38%)	83 (4.75%)
Val	523 (81.85%)	54 (8.45%)	32 (5.01%)	30 (4.69%)
Test	497 (80.94%)	56 (9.12%)	33 (5.37%)	28 (4.56%)

V. EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

A. Evaluated Model Architectures

We evaluate three state-of-the-art nano-scale real-time object detectors:

- 1) **Ultralytics YOLO11n** [1]: Compact convolutional single-stage detector featuring C3k2 and SPPF modules (2.6M parameters). Loss function optimizes Complete IoU (CIoU), Binary Cross-Entropy (BCE), and Distribution Focal Loss (DFL):

$$\mathcal{L}_{\text{YOLO11}} = \lambda_{\text{box}} \mathcal{L}_{\text{CIoU}} + \lambda_{\text{cls}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{dfl}} \mathcal{L}_{\text{DFL}} \quad (4)$$

- 2) **Ultralytics YOLO26n** [2]: Next-generation NMS-free end-to-end detector optimizing edge throughput (2.4M parameters) via dual-label one-to-one assignment.
- 3) **D-FINE-N** [3]: Lightweight Real-Time Detection Transformer with HGNetv2 backbone and Fine-grained Distribution Refinement (3.8M parameters) optimized via Hungarian set loss:

$$\mathcal{L}_{\text{D-FINE}} = \lambda_{\text{cls}} \mathcal{L}_{\text{VFL}} + \lambda_{\text{box}} \mathcal{L}_{\text{GIoU}} + \lambda_{\text{fdr}} \mathcal{L}_{\text{FDR}} + \mathcal{L}_{\text{GO-LSD}} \quad (5)$$

B. Controlled Training Controls

All models are fine-tuned under strictly locked shared controls:

- **Fine-Tuning Budget**: Exactly 220 epochs without early stopping.

DMS-Eval: End-to-End Benchmark Framework & Evaluation Protocol

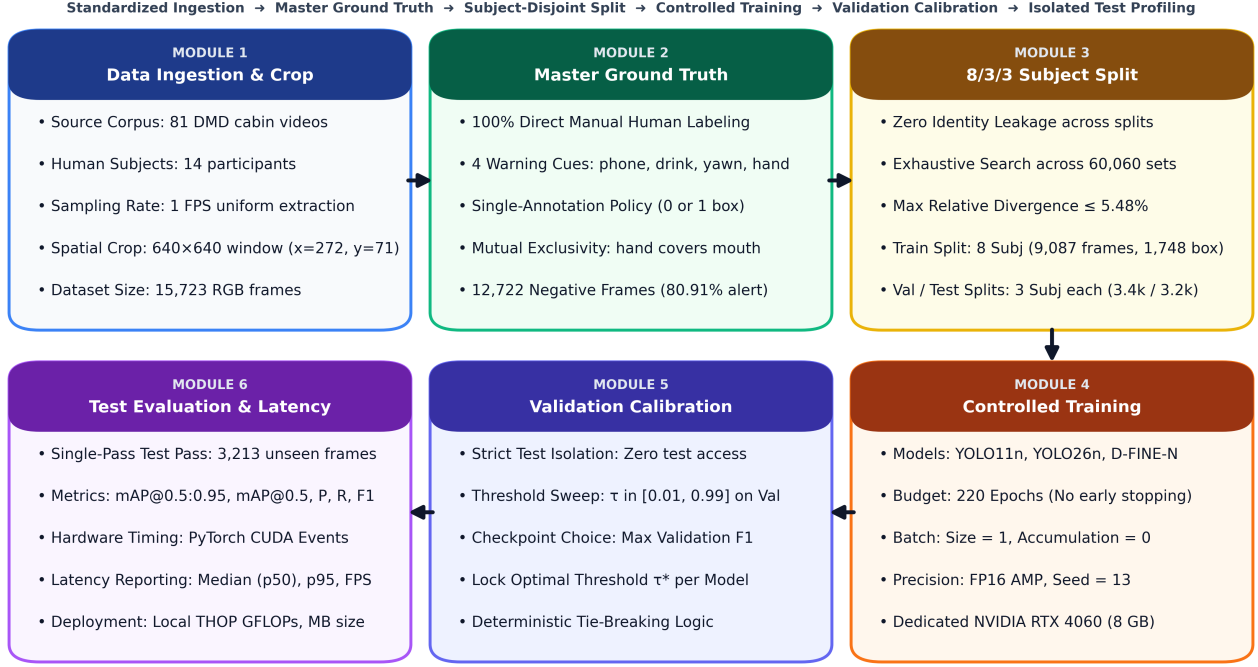


Fig. 4. DMS-Eval end-to-end benchmark framework and evaluation lifecycle: spanning standardized 640×640 frame extraction from 81 DMD videos, authoritative Label Studio ground truth annotation across 4 target visual warning cues (80.91% negative frames), deterministic 8/3/3 subject-disjoint split optimization ($\leq 5.48\%$ relative divergence), controlled training on RTX 4060 hardware (batch size 1, 220 epochs, seed 13), validation-only F_1 score threshold calibration (τ^*), and isolated single-pass test evaluation.

- **Batch Size:** Batch size = 1 (gradient accumulation disabled) to emulate real-time edge frame ingestion.
- **Deterministic Seed:** Random seed fixed to 13 across PyTorch, CUDA, and CuDNN.
- **Precision:** FP16 Automatic Mixed Precision (AMP).
- **Hardware Platform:** Dedicated NVIDIA GeForce RTX 4060 GPU (8 GB VRAM).

Model-specific optimizer configurations (SGD vs. AdamW), learning rate schedules, and data augmentations follow each architecture’s official repository recipe to ensure peak performance without unfair hyperparameter distortion.

C. Validation Calibration and Test Isolation

To prevent data snooping, model checkpoints and confidence thresholds (τ) are selected strictly on the Validation split. For each model, we sweep confidence thresholds $\tau \in [0.01, 0.99]$ with a step of 0.01 and identify the threshold τ^* that maximizes the micro-averaged F_1 score:

$$\tau^* = \arg \max_{\tau \in [0.01, 0.99]} F_1(\tau; S_{\text{val}}) = \frac{2 \cdot P(\tau) \cdot R(\tau)}{P(\tau) + R(\tau)} \quad (6)$$

where $P(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FP}(\tau)}$ and $R(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FN}(\tau)}$ under COCO one-to-one IoU ≥ 0.50 matching. The optimal validation checkpoint and threshold τ^* are permanently frozen.

Final benchmark evaluation is executed in a single pass on the unseen Test split (S_{test}) without further tuning.

D. Latency and Footprint Profiling

In accordance with standard edge ML benchmarking guidelines [15]–[17], inference speed is measured under native PyTorch in FP16 on the RTX 4060 GPU at batch size 1. Following 10 untimed warm-up passes, inference latency is measured across all test frames using hardware-synchronized CUDA events (`torch.cuda.Event`). We report median latency (p_{50}), 95th percentile (p_{95}), 99th percentile (p_{99}), throughput (FPS), parameter count, and computational complexity in GFLOPs.

VI. BENCHMARK RESULTS FRAMEWORK

VII. DISCUSSION AND THREATS TO VALIDITY

A. Architectural Trade-offs at Edge Scale

Evaluating nano-scale detectors reveals crucial trade-offs between inductive bias and global receptive field. While convolutional backbones (YOLO11n, YOLO26n) benefit from strong spatial locality, Detection Transformers (D-FINE-N) leverage cross-scale attention to disambiguate occluded interactions. However, transformer query projections incur higher on-chip SRAM bandwidth demands, highlighting the need

TABLE III
CONTROLLED COMPARATIVE EVALUATION RESULTS FRAMEWORK (RTX 4060, BATCH SIZE 1, FP16)

Model Architecture	Params (M)	FLOPs (G)	Latency p50 (ms)	Throughput (FPS)	mAP@0.5:0.95	mAP@0.5	Precision	Recall	F1 Score
Ultralytics YOLO11n	2.6M	6.5G	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]
Ultralytics YOLO26n	2.4M	5.8G	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]
D-FINE-N	3.8M	8.4G	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]

TABLE IV
PER-CLASS DETECTION PERFORMANCE FRAMEWORK ($mAP@0.5 : 0.95$
/ $mAP@0.5$ ON TEST SPLIT)

Model	phone_use	drinking	yawning	hand_mouth
YOLO11n	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]
YOLO26n	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]
D-FINE-N	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]	[TO_BE_FILLED]

to balance mAP gains against embedded memory constraints [17].

B. Threats to Validity

- **Internal Validity:** Fixed training seeds and locked optimizer configurations guarantee fair comparisons. Validation-only threshold calibration prevents optimistic test leakage.
- **External Validity:** Performance may vary under extreme nighttime illumination or across unobserved camera mounting angles.

VIII. CONCLUSION

This paper introduced **DMS-Eval**, a rigorous empirical benchmark for evaluating lightweight object detectors on in-cabin driver monitoring tasks. By establishing an authoritative 15,723-frame dataset, four clearly defined warning cues, and an optimal 8/3/3 subject-disjoint split, DMS-Eval eliminates identity leakage and ensures reproducible evaluation. The controlled comparison between YOLO11n, YOLO26n, and D-FINE-N provides crucial guidance for deploying robust, real-time driver assistance systems on resource-constrained automotive hardware.

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